

Project Plan & Management Specification

Operational Roadmap, WBS, Milestones, RACI Assignment, Risk Mitigation, and KPIs — End-to-End Business Analytics Platform

Initiative / Track:	DEPI — AI & Data Science Track	Execution Period:	October 06, 2026 – December 04, 2026
Methodology:	Hybrid Agile/Scrum & CRISP-DM Framework	Document ID:	DEPI-PLN-2026-V1.0
Deliverable Focus:	Project Planning & Management (DEPI Section 1)	Repository Target:	GitHub Portfolio Organization

1. Project Management Methodology & Governance

To ensure the highest degree of engineering quality, reproducibility, and deadline adherence, the project integrates the domain-specific **CRISP-DM** (Cross-Industry Standard Process for Data Mining) lifecycle with an iterative **Agile / Scrum** operational framework.

- Sprint Cadence & Rituals:** Execution is organized into four two-week sprints. Each sprint begins with Backlog Grooming and Sprint Planning, incorporates twice-weekly asynchronous standups on GitHub Discussions, and concludes with a Sprint Review and Retrospective.
- Version Control & Branching Strategy:** The project adheres to a strict GitFlow branching model. The main branch contains production-ready, passing code. Active development takes place on `develop`, with isolated branches for features (`feature/data-ingest`, `feature/ml-modeling`, `feature/dashboard-ui`). Merges into `develop` require a code review and passing automated CI tests.
- Code Review & Quality Gates:** Pull requests require peer review from at least one core engineer. Linting (Flake8), automated testing (Pytest), and type validations are executed automatically via GitHub Actions on every pull request before merging is permitted.
- Documentation & Reproducibility Standards:** All modules, classes, and endpoints follow Google Python Style docstrings, with auto-generated OpenAPI/Swagger specifications for API endpoints and reproducible markdown summaries for analytical notebooks.

2. Work Breakdown Structure (WBS) & DEPI Milestones

The Work Breakdown Structure maps directly to the official DEPI evaluation timeline, translating high-level objectives into verifiable milestones:

ID	Milestone Name	Official DEPI Due	Key Work Packages	Core Deliverables
M1	Planning & Governance	16 Oct 2026	Team charter, repository setup, scope boundary definition, risk assessment.	Project Proposal, Project Plan, GitHub Repo.
M2	Literature & Tech Review	16 Oct 2026	Survey of analytics architectures, ML benchmarks, and BI dashboard design.	Literature Review Document, Benchmark Summary.

ID	Milestone Name	Official DEPI Due	Key Work Packages	Core Deliverables
M3	Requirements Gathering	16 Oct 2026	Stakeholder interviews, user stories, functional and non-functional requirements.	SRS Document, Use Case Catalog, Traceability Matrix.
M4	System Analysis & Design	06 Nov 2026	Architecture diagram, ERD database schema, DFD data flow, UI wireframes.	System Design Document, ERD, DFD, Wireframes.
M5	Implementation & ML	30 Nov 2026	Multi-source ingestion, ETL pipeline, statistical EDA, ML modeling, FastAPI, Docker.	Source Code, EDA Notebooks, Model Binaries, Dashboard.
M6	Testing, Reports & Demo	04 Dec 2026	Unit/integration testing, bug logging, user manual, final presentation, infographic.	Test Reports, User Guide, Presentation, Infographic.

3. Master Project Schedule & Gantt Timeline

The project spans 9 active weeks from early October to early December 2026. The table below represents the phased Gantt distribution:

Phase / Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9
Phase 1: Planning & Charters	■	■							
Phase 2: Literature & Requirements	■	■							
Phase 3: System Analysis & ERD		■	■	■	■				
Phase 4: Data Pipelines & Ingestion			■	■	■				
Phase 5: EDA & Executive Dashboard				■	■	■	■		
Phase 6: ML Modeling & Validation					■	■	■	■	
Phase 7: FastAPI, Docker & CI/CD						■	■	■	
Phase 8: Testing, QA & Documentation							■	■	■
Phase 9: Final Demo & Evaluation								■	■

Note: Weeks W1–W2 (Oct 06–16), W3–W5 (Oct 17–Nov 06), W6–W8 (Nov 07–30), W9 (Dec 01–04, 2026).

4. Resource Allocation & Technical Stack

The technical environment is selected for production readiness, open-source compliance, and seamless cross-platform deployment:

Layer / Domain	Selected Technology Stack	Purpose & Operational Rationale
Data Ingestion	PostgreSQL, Python requests, BeautifulSoup4, Selenium	Connectors for relational SQL data, external REST APIs, and public web scraping.
Data Engineering	Pandas, NumPy, SQLAlchemy, Pydantic	Data cleaning, schema validation, type enforcement, and staging warehouse ETL.
Analytics & EDA	JupyterLab, Matplotlib, Seaborn, SciPy	Statistical profiling, distribution fitting, correlation, and feature exploration.
BI Dashboard	Streamlit, Plotly Express	Interactive web application for executive KPI tracking, drill-downs, and maps.
Machine Learning	Scikit-Learn, XGBoost, LightGBM, MLflow	Multi-model training (Regression, Classification, Clustering) and experiment tracking.
Serving & Deployment	FastAPI, Uvicorn, Docker, Docker-Compose	Lightweight, asynchronous REST API serving real-time model predictions.
DevOps & CI/CD	GitHub Actions, Pytest, Flake8	Automated linting, unit/integration testing, and Docker container build verification.

5. Team Roles & Task Assignment (RACI Matrix)

Governance is established through defined functional roles and a clear **RACI Matrix** (Responsible, Accountable, Consulted, Informed):

Project Phase & Deliverables	Project Lead	Data Eng.	BI Analyst	ML Eng.	DevOps Eng.
Project Proposal & Management Plan	A / R	C	C	C	C
Literature Review & Tech Survey	A	C	R	R	I
Requirements & User Stories (SRS)	A / R	C	R	C	I
System Architecture & ERD Schema	A	R	C	C	R
Data Extraction (SQL / API / Scraper)	A	R	C	I	I
ETL, Data Cleaning & Feature Staging	A	R	R	C	I
Exploratory Data Analysis (EDA)	A	I	R	R	I
Executive BI Dashboard (Streamlit)	A	I	R	C	I
ML Model Training & Benchmarking	A	I	C	R	C
FastAPI Endpoint & Containerization	A	I	I	R	R

Project Phase & Deliverables	Project Lead	Data Eng.	BI Analyst	ML Eng.	DevOps Eng.
Testing, QA & Bug Fixing	A	R	R	R	R
Final Documentation & Presentation Deck	A / R	R	R	R	R

6. Risk Assessment & Mitigation Strategy

The risk management plan identifies critical vulnerabilities across technical and operational dimensions:

Risk Domain	Identified Risk & Trigger	Severity / Likelihood	Mitigation Protocol & Contingency Plan
Data Ingestion	Third-party REST API rate-limiting or anti-scraping IP blocks.	High / Med	Implement retry logic with exponential backoff, persistent local SQLite caching, and maintain offline fallback datasets.
Data Quality	High missingness (>30%) or inconsistent data types in scraping tables.	High / High	Enforce Pydantic schema validation at ingestion; establish automated imputation pipelines and drop corrupt records automatically.
Model Performance	Overfitting on training data or poor generalization on test sets.	High / Med	Enforce stratified 5-fold cross-validation, apply regularization (L1/L2), conduct feature importance pruning, and log all runs in MLflow.
Deployment	Dependency version conflicts or heavy Docker container build sizes.	Med / Med	Lock package versions via Poetry/pip-tools, utilize multi-stage minimal Python-slim base images, and verify locally prior to push.
Scope & Schedule	Feature creep pushing ML and deployment beyond the Nov 30 deadline.	High / Low	Adhere strictly to MoSCoW prioritization; mark cloud hosting and advanced MLflow as optional extensions if core deliverables lag.

7. Key Performance Indicators (KPIs)

Project evaluation will be measured across four rigorous KPI categories:

KPI Category	Performance Indicator	Target Threshold	Measurement Method
System Performance	FastAPI Inference Latency	P95 < 200 ms	Automated benchmark using Apache Benchmark / Locust.
	Pipeline Execution Duration	< 15 minutes	End-to-end execution timer for SQL + ETL jobs.
Machine Learning	Classification Performance	F1-Score ≥ 0.82	Cross-validated test set evaluation on held-out data.
	Regression Performance	R² ≥ 0.80, RMSE < baseline	Cross-validated evaluation on continuous target.
	Clustering Quality	Silhouette Score ≥ 0.50	Cluster cohesion and separation metrics.

KPI Category	Performance Indicator	Target Threshold	Measurement Method
Software Quality	Automated Test Coverage	≥ 80% line coverage	Pytest test runner with Coverage.py.
	Code Quality & Cleanliness	Zero fatal Flake8 lint errors	GitHub Actions automated lint check.
Business Impact	Report Generation Velocity	> 75% time reduction	Comparison between manual spreadsheet vs dashboard.

8. Communication & Progress Reporting Protocol

Structured communication channels ensure transparency across all project contributors and academic supervisors:

- **Weekly Status Reports:** Published every Sunday to the GitHub Project Wiki, detailing completed tasks, blockers, and next week priorities.
- **Sprint Retrospectives:** Conducted bi-weekly to assess team velocity, adjust task distributions, and resolve emerging friction points.
- **Supervisor Check-ins:** Aligned with major DEPI submission dates (Oct 16, Nov 06, Nov 30, Dec 04) for formal review and grading.
- **Deliverable Archiving:** All finalized documents, slides, and architectural diagrams are mirrored to the project GitHub repository under /docs.

9. Quality Assurance & Final Sign-Off

This operational plan has been formulated in direct compliance with the Digital Egypt Pioneers Initiative (DEPI) Capstone Documentation Guidelines. All technical work packages, risk controls, and timeline gates are committed to by the multidisciplinary project team to guarantee exemplary execution and academic excellence.